Notes: Rajan (JF, 1992)

Insiders and Outsiders: The Choice between Informed and Arm's-Length Debt

Contents

1 Big picture

- **Question.** If banks are better monitors and better informed than dispersed public creditors, why do many firms diversify away from bank financing and issue arm's-length debt?
- **Core answer.** Bank finance has a benefit and a cost. The benefit is *control*: an informed bank can react to inside information and stop inefficient continuation. The cost is *hold-up*: once the project is under way, the informed bank can bargain for part of the project's surplus.
- **Setting.** An owner-managed firm must borrow to fund a project. After investment, the owner chooses costly effort. At the interim date, the owner and an inside bank learn whether the project is in a good or bad state, while outside lenders do not.
- **Main friction.** Contracts cannot be conditioned on effort, the state, or the continuation decision. This means the ex post allocation of control is partly determined by who is informed and who can renegotiate.
- **Main trade-off.** Arm's-length debt leaves the owner with stronger incentives because outsiders cannot easily renegotiate away the good-state surplus. But outsiders also cannot prevent continuation of negative-NPV projects. Bank debt fixes the continuation problem, but the bank's informational advantage gives it bargaining power that weakens the owner's ex ante effort incentives.
- **Maturity result.** Short-term bank debt and long-term bank debt distort effort in *different states*. With short-term bank debt, bargaining occurs in the good state because the owner needs rollover finance. With long-term bank debt, bargaining occurs in the bad state because the bank must persuade the owner to liquidate. The optimal maturity depends on where bargaining is least damaging.
- **Quality result.** High-quality firms tend to prefer arm's-length debt because control is less valuable for them and bank rent extraction becomes relatively more important. Lower-quality firms can prefer bank finance because avoiding inefficient continuation matters more.
- **Competition result.** Once outside lenders can compete at the interim date, the inside bank is no longer perfectly locked in. Its rents and its control both become endogenous outcomes of a winner's-curse bidding game.
- **Capital-structure result.** Firms can use *multiple sources* of borrowing and the *priority structure* of claims to reduce bank rents while preserving bank control. In the model, arm's-length debt should optimally have priority over bank debt at the final date.
- **Signal result.** More informative interim public signals can make bank finance more attractive, because they reduce the inefficiencies of outside lending and improve the control environment. This helps explain why bank debt may be used early in a project and replaced later by public debt.
- **Economic takeaway.** The cost of informed lending is not mainly the explicit interest rate. It is the lender's ex post bargaining power created by private information.

2 Setup and economic objects (key objects)

2.1 Project, timing, and real payoffs

- At date 0 the owner must invest a fixed amount I.
- After investment, the owner exerts effort β at unit cost 1.
- At date 1 the project can be liquidated for value L, where $L < I$.
- At date 2 the assets are worthless if liquidation did not already occur.
- The date-1 state is either **good** (G) or **bad** (B).
- In the good state the project pays X for sure at date 2.
- In the bad state the project pays X with probability p_B and 0 with probability $(1 - p_B)$.
- The paper assumes
$$X > I > L > p_B X.$$

Interpretation. Continuation is efficient in the good state and inefficient in the bad state. If the owner had unlimited ability to commit, she would always continue in G and liquidate in B.

2.2 Effort and project quality

- The probability of the good state is $q(\beta, \theta)$.
- $\theta \in [0, 1]$ is an exogenous quality parameter: higher θ means a better project.
- Effort raises the success probability but at a diminishing rate:
$$q_1(\beta, \theta) > 0, \qquad q_{11}(\beta, \theta) < 0, \qquad q_2(\beta, \theta) > 0.$$
- The cross-partial is set to zero in the baseline, so effort and quality enter separably.

Interpretation. The owner can improve the probability of landing in the good state, but only by bearing a private effort cost. This is where financing affects real behavior: debt choice changes the owner's incentive to exert effort.

2.3 Information structure

- At date 0 everyone observes project quality θ.
- After lending begins, the owner privately knows the effort exerted.
- At date 1 the owner learns the state before deciding whether to continue.
- An inside bank that lent at date 0 also learns the effort and the state through monitoring and access to the firm's records.
- Arm's-length lenders and outside banks observe only public information, which is initially taken to be uninformative.

Interpretation. The inside bank's advantage is not just that it is present. It is that it possesses soft information the firm cannot credibly transmit to outsiders.

2.4 Financial contracts and lender types

- The model allows only debt contracts.
- The firm can borrow A_i at date i and promise a single repayment D_{ij} at date j.
- If $j - i = 1$, the contract is **short-term**; if $j - i = 2$, it is **long-term**.
- A **bank** can become informed by lending and can renegotiate easily because it is a concentrated creditor.
- An **arm's-length lender** does not become informed and is hard to coordinate with ex post.

2.5 Key assumptions doing the heavy lifting

- Contracts cannot be made contingent on effort, the state, or the liquidation decision.
- The date-0 credit market is competitive, so lenders break even in expectation.
- Any ex post rents lenders expect to capture are competed away up front and rebated to the owner at date 0.
- Initially the firm chooses only one financing mode at a time, and if it borrows from a bank it is locked in to that bank for future borrowing. Both assumptions are later relaxed.

Interpretation. Because all loans are zero NPV at date 0, the true cost of a financing choice is not the expected transfer to lenders. It is the *behavioral distortion* induced in effort and continuation.

3 Model and the basic trade-off (all equations + interpretation)

3.1 First-best benchmark

If the project is financed and continuation is chosen efficiently, expected surplus is

$$q(\beta,\theta)(X-I) - (1-q(\beta,\theta))(I-L) - \beta.$$

The first-best effort level solves

$$q_1(\beta,\theta) = \frac{1}{X-L}, \qquad \beta = \beta^*_{FB}. \tag{1}$$

Interpretation.

- The marginal benefit of effort is the extra probability of reaching the good state times the incremental value of being in G rather than B, which is $X - L$.
- Any financing arrangement is evaluated against this benchmark: does it keep continuation efficient, and does it preserve the owner's incentive to supply effort?

3.2 Arm's-length debt: no control, weak continuation discipline

Under arm's-length lending, the owner borrows I at date 0 and promises D_{02a} at date 2. Because outside creditors neither observe the state nor coordinate well, the owner continues even in the bad state. Her effort problem is

$$\max_{\beta} \; q(\beta,\theta)(X - D_{02a}) + (1 - q(\beta,\theta))p_B(X - D_{02a}) - \beta. \tag{2}$$

Lender individual rationality requires

$$D_{02a} \geq \frac{I}{q(\beta_a^c,\theta) + (1 - q(\beta_a^c,\theta))p_B}, \tag{3}$$

and rational expectations requires

$$\beta_a^c = \beta_a^*. \tag{4}$$

Combining the owner's FOC, lender break-even, and rational expectations yields

$$q_1(\beta,\theta) = \frac{1}{\left[X - I - \dfrac{(1 - q(\beta_a^*,\theta))(1 - p_B)I}{q(\beta_a^*,\theta) + (1 - q(\beta_a^*,\theta))p_B}\right](1 - p_B)}, \qquad \beta = \beta_a^*. \tag{5}$$

Interpretation.

- Arm's-length creditors cannot stop inefficient continuation in the bad state.
- Because the owner retains some upside even in the bad state through limited liability, bad states become less painful. This weakens effort.
- The lender responds by charging a higher face value, which further reduces good-state surplus and can trigger credit rationing for low-quality projects.

3.3 Short-term bank debt: efficient control, but hold-up in the good state

Suppose the bank lends short-term. It is repaid at date 1. In the bad state, the informed bank can refuse rollover and force liquidation, which is efficient. In the good state, the owner needs fresh credit to continue, so the bank can bargain over the continuation surplus. The owner gets $\mu(X - L)$ and the bank gets $(1 - \mu)(X - L) + L$.

The owner's effort FOC is

$$q_1(\beta,\theta) = \frac{1}{\mu(X - L)}, \qquad \beta = \beta_{Sb}^*, \tag{6}$$

and bank participation requires

$$\left[1 - \frac{I - L}{q_{Sb}^*(X - L)}\right] - \mu \geq 0. \tag{7}$$

Interpretation.

- Short-term bank debt gives the bank exactly the kind of control arm's-length lenders lack: it can stop negative-NPV continuation.

- But the price of that control is hold-up in the good state. The owner no longer keeps the full good-state surplus.
- Higher owner bargaining power μ raises effort because it lets the owner keep more of the upside, but if μ is too high the bank may not lend because it cannot recover expected losses from bad states.

3.4 Long-term bank debt: no hold-up in the good state, but bargaining in the bad state

Now suppose the bank lends long-term at date 0 and is repaid only at date 2. Then there is no need to renegotiate in the good state. However, if the state is bad, efficient liquidation now requires bargaining because the bank cannot unilaterally force repayment at date 1. The bank must, in effect, bribe the owner to stop.

The break-even face value for long-term bank debt is

$$X \geq \frac{I - (1 - q^*_{Lb})(1 - \mu)(L - p_B X)}{q^*_{Lb} + (1 - q^*_{Lb})p_B} = D_{02b}. \tag{8}$$

Interpretation.

- Long-term bank debt removes the good-state hold-up problem.
- But it creates a new incentive problem: the owner now expects to receive part of the liquidation surplus in bad states, so bad states become less painful.
- This can reduce effort even more when the owner's bargaining power is high, because she is partly insured against the downside.

3.5 Ex ante utilities and the bank-versus-market choice

Because date-0 competition rebates expected rents to the owner, ex ante utility under bank finance is

$$q^*_b(X - I) - (1 - q^*_b)(I - L) - \beta^*_b. \tag{9}$$

Under arm's-length finance it is

$$q^*_a(X - I) - (1 - q^*_a)(I - p_B X) - \beta^*_a. \tag{10}$$

A bank loan is preferred when

$$(1 - q^*_a)(L - p_B X) - (q^*_a - q^*_b)(I - L) - \Big[(q^*_a - q^*_b)(X - I) - (\beta^*_a - \beta^*_b)\Big] \geq 0. \tag{11}$$

Interpretation of equation (11).

- The first term is the **benefit of bank control**: with bank debt, the project is liquidated in bad states instead of being inefficiently continued.
- The second and third terms are the **cost of bank finance**: lower effort means bad states occur more often and total expected surplus is lower.
- So the choice is not "banks good, markets bad" or the reverse. It is a clean efficiency comparison between *better continuation decisions* and *worse ex ante incentives*.

3.6 What bargaining power does to maturity choice

The paper derives two threshold results.

- As owner bargaining power μ rises, short-term bank debt becomes more attractive because the good-state hold-up becomes less severe.
- As μ rises, long-term bank debt becomes less attractive because the owner captures more of the bad-state liquidation surplus, which further weakens effort.

Proposition 1.

- If the short-term and long-term thresholds are ordered in the usual way, the firm chooses **short-term bank debt** for high μ, **arm's-length debt** for intermediate μ, and **long-term bank debt** for low μ.
- If the thresholds overlap the other way, the firm may never choose arm's-length debt and instead switch directly between short-term and long-term bank borrowing.

Interpretation. Maturity is being used to determine *which state* becomes the bargaining state. Short maturity puts bargaining in the good state; long maturity puts bargaining in the bad state.

3.7 What project quality does to the financing choice

Proposition 2. There exists a quality cutoff $\bar{\theta}$ such that low-quality projects prefer short-term bank debt while high-quality projects prefer arm's-length debt.

Interpretation.

- When quality is low, continuation mistakes are expensive, so bank control is valuable.
- When quality is high, the public-debt face value falls and the market-financing effort distortion shrinks. Bank rent extraction then becomes relatively more important.
- This gives a financing life-cycle logic: more opaque or weaker projects rely more on informed lenders; better projects move toward arm's-length debt.

4 Interim competition, multiple sources, and priority

4.1 Interim competition and the winner's curse

Rajan next relaxes exogenous lock-in. At date 1, the owner can ask both the inside bank and an outside bank to bid for rollover financing. To simplify the algebra he assumes $p_B = 0$ and $L = I$, so the first-period loan is riskless and the amount that must be rolled over is $I_S = I$.

Proposition 3.

- There is *no pure-strategy equilibrium* in the interim bidding game.
- There is a mixed-strategy equilibrium in which the outside bank sometimes stays out and otherwise bids without conditioning on the state.

If the outside bank's conjectured success probability is q, the probability it does *not* bid is

$$\pi_N = \frac{1-q}{q}\frac{I_S}{X - I_S}, \qquad \text{more precisely } \pi_N = \min\left\{1, \frac{1-q}{q}\frac{I_S}{X-I_S}\right\}. \tag{13}$$

Conditional on the good state, the outside bank's expected profit is

$$\frac{1-q}{q}\, I_S(1-\pi_N), \tag{12}$$

and the inside bank's expected profit is

$$\frac{(1-q)I_S}{q}. \tag{14}$$

Interpretation.

- If the outside bank used a predictable pure strategy, the inside bank would respond by lending only in good states, leaving the outsider with only lemons. That is the winner's-curse logic.
- The mixed strategy lets outsiders sometimes finance the bad state, but only in a way that makes them break even on average.
- The inside bank now controls continuation only with probability π_N, not with certainty. So interim competition simultaneously lowers the bank's rents and its control.

4.2 Comparative statics of inside-bank power

Proposition 4.

- Both inside-bank rents and inside-bank control decrease with project quality q.
- Both increase with the amount that must be rolled over at date 1, I_S.
- Control increases when the uncommitted surplus $X - I_S$ becomes smaller.

Interpretation. The inside bank's power is really *implicit property rights*. It is strongest when outsiders face the worst information problem and when the firm's need for refinancing is largest.

4.3 Why firms borrow from both banks and markets

Once inside-bank power depends on how much refinancing is needed at the interim date, the firm can strategically weaken the bank by reducing the amount that must be rolled over with it. Borrowing some funds from arm's-length lenders at date 0 does exactly that.

- Less bank debt at date 0 means less has to be refinanced by the inside bank at date 1.
- That makes outside bidding more credible and reduces the inside bank's informational monopoly.
- The downside is that weaker bank monopoly can also mean weaker bank control.

Interpretation. The model gives a rational basis for a mixed financing structure. Firms are not combining sources because they are indifferent; they are combining sources to trade off bank control against bank rent extraction.

4.4 Priority structure and Proposition 5

The key design question is which creditor gets priority over date-2 revenues.

Proposition 5. Arm's-length debt contracted at date 0 should optimally have a *prior claim* over date-2 revenues relative to bank debt.

Why?

- If arm's-length debt is prior, less date-2 surplus remains uncommitted for the inside bank and the owner to bargain over at date 1.
- Lower uncommitted surplus reduces outsiders' willingness to bid aggressively against the inside bank, which raises the bank's control.
- At the same time, the arm's-length lender becomes safer because it is protected by priority, so it charges a lower promised repayment.

Economic interpretation.

- The model predicts that public debt should often be secured on hard assets or otherwise be effectively prior, while bank debt can be junior or backed by assets whose value is closely tied to firm continuation.
- Priority is therefore not just about default recovery. It is also about shaping the ex post bargaining game between informed and uninformed lenders.

4.5 Interim public information and Proposition 6

Now suppose outsiders receive a noisy public signal at date 1. If the project is in the good state, signal W_G arrives with probability Ω_G; if it is in the bad state, signal W_B arrives with probability Ω_B, where the signal is informative but noisy.

The outside bank updates its prior q^* to

$$q^+ = \frac{q^*\Omega_G}{q^*\Omega_G + (1-q^*)(1-\Omega_B)}. \tag{15}$$

If the bad signal arrives, the posterior is

$$q^- = \frac{q^*(1-\Omega_G)}{q^*(1-\Omega_G) + (1-q^*)\Omega_B}. \tag{16}$$

Main result. More informative interim public signals can increase the ex ante advantage of bank debt.

Interpretation.

- A better public signal makes outsider lending more state contingent.
- That improves continuation discipline and can raise the owner's effort.
- This yields a natural maturity pattern: early bank borrowing when information is still soft, later arm's-length borrowing when credible public information has emerged.

5 What makes the argument credible (and where it is limited)

5.1 What does the heavy lifting

- **Soft information.** Banks learn information through lending that firms cannot easily communicate to outsiders.
- **Renegotiation asymmetry.** A concentrated bank can renegotiate; dispersed public creditors cannot.
- **Limited contractibility.** Contracts cannot specify efficient continuation directly, so control is allocated indirectly through financing structure.
- **Competitive date-0 markets.** This strips away simple transfer arguments and isolates the real efficiency margins: effort and continuation.

5.2 Why the theory is disciplined rather than ad hoc

- The first-best benchmark is explicit.
- Each contract is evaluated using the same objects: continuation efficiency, lender break-even, and induced effort.
- The financing predictions are comparative-static results, not loose stories: bargaining power, project quality, interim competition, priority, and public signals each move the optimum in a definite direction.

5.3 What the paper explains well

- Why banks are useful even when market lenders are competitively priced.
- Why firms may *voluntarily* move away from banks despite the monitoring benefits.
- Why debt maturity matters even without standard term-premium stories.
- Why firms often use multiple creditors and carefully designed priority structures.
- Why bank debt is common in early project stages and market debt becomes more attractive once information becomes public.

5.4 Main limitations

- **Debt only.** The baseline excludes equity and richer contingent contracts.
- **Owner-managed firm.** The model abstracts from internal agency conflicts between managers and shareholders.
- **Perfect inside-bank information.** Monitoring is effectively costless and very accurate.
- **Exogenous bargaining power in the first half.** The basic maturity results rely on a reduced-form bargaining parameter μ before interim competition is endogenized.

- **Strong simplifications in Section III.** The mixed-strategy analysis sets $p_B = 0$ and $L = I$ to make the bidding game tractable.
- **No full mechanism-design solution.** The paper discusses loan commitments and regulation, but the main results are derived in incomplete-contract environments.

5.5 Relation to the literature

- The paper complements Diamond's work on bank monitoring by emphasizing that informed lending creates *ex post surplus extraction*, not just ex ante screening benefits.
- It is closely related to incomplete-contract and property-rights thinking: the allocation of control depends on who can intervene when contingencies are unverifiable.
- Its priority result differs from models where the main bank problem is excessive liquidation. Here the informed bank's issue is *too much bargaining power*, not too little.

6 Figures 1–4 and key theoretical results

6.1 Figure 1: effort under bank debt versus arm's-length debt

Read.

- The figure compares the owner's effort when the project is financed entirely by arm's-length debt and when it is financed entirely by short-term bank debt.
- As project quality rises, the two effort levels move closer together.

Why it matters.

- This is the paper's cleanest illustration that the cost of bank finance is strongest when projects are weak and opaque.
- For better projects, bank control matters less and the effort distortion from bank bargaining becomes small.

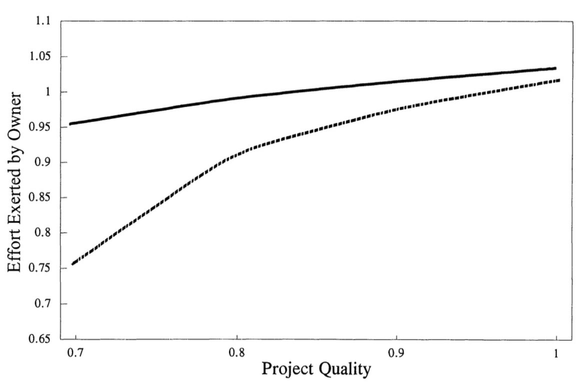

Figure 1. The effect of project quality on the effort exerted by the owner. The solid line is the effort exerted by the owner when funded entirely with arm's-length debt while the dashed line is the effort exerted when funded entirely with short-term bank debt. Calculations are based on the data in example 1.

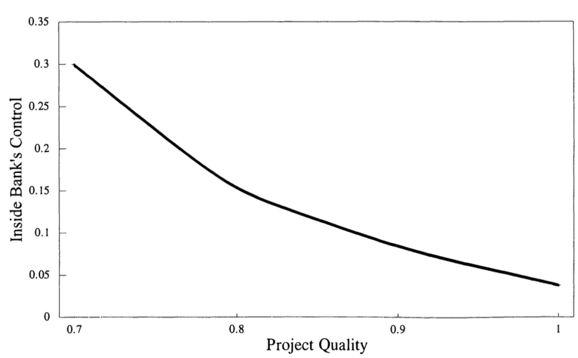

Figure 2. The effect of project quality on the control provided by the inside bank. The control provided by the inside bank is the ex ante probability that the outside bank will not bid. Calculations are based on the data in example 1.

6.2 Figure 2: inside-bank control falls with quality

Read.

- The figure plots the inside bank's control, measured by the probability π_N that the outside bank does not bid.
- Control declines as project quality improves.

Why it matters.

- When projects are better, outsiders are less worried about adverse selection, so they compete more aggressively.
- That erodes the inside bank's implicit monopoly and reduces both its rents and its influence.

6.3 Figure 3: who should finance the project?

Read.

- The figure plots the owner's ex ante utility under arm's-length debt minus the utility under short-term bank debt.
- The gap is positive for low-quality projects, negative for medium-quality projects, and approaches zero again for high-quality projects in the numerical example.

Why it matters.

- Bank debt is not uniformly best for opaque firms and market debt is not uniformly best for transparent firms.
- In the interim-competition environment, the main role of banks is strongest in the middle region where control has large value but hold-up is not too severe.

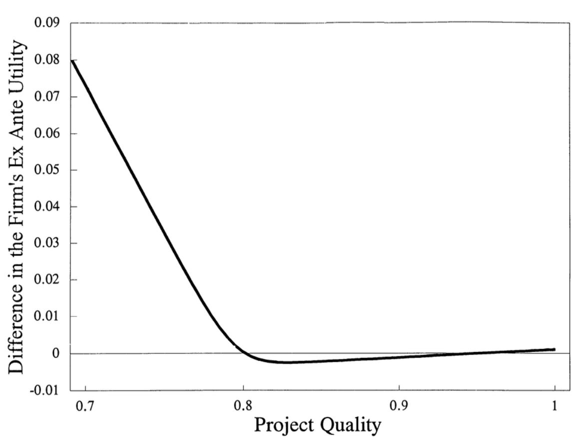

Figure 3. The effect of project quality on the difference in ex ante value to the owner of the different sources. The owner's ex ante expected utility when funded entirely with short-term bank debt is subtracted from the ex ante expected utility when funded entirely with arm's-length debt. This difference is plotted in the figure. Calculations are based on the data in example 1.

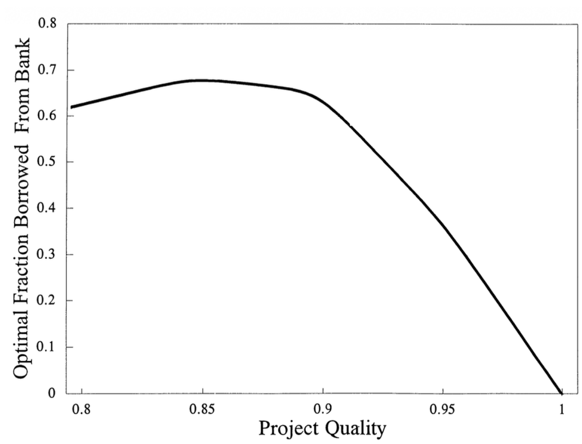

Figure 4. The effect of project quality on the optimal fraction of investment funded with short-term bank debt. The optimal fraction of short-term bank debt is calculated, assuming the arm's-length lender has priority. Calculations are based on the data in example 1.

6.4 Figure 4: optimal bank share declines with quality

Read.

- The figure plots the optimal fraction of date-0 investment financed with short-term bank debt when arm's-length lenders have priority.
- The optimal bank fraction falls as project quality rises.

Why it matters.

- This is the paper's structural prediction for mixed financing.
- Better projects need less informed control and therefore rely less on bank finance.
- The figure ties together the paper's three big messages: bank debt is useful, bank debt is costly, and capital structure is a tool for balancing the two.

7 Short summary

- **Method.** Rajan builds an incomplete-contract model in which banks obtain soft information by lending and can renegotiate efficiently, while arm's-length lenders remain uninformed and dispersed.
- **Main conceptual result.** The value of bank finance comes from control, but the cost comes from ex post bargaining power over the project's surplus.
- **Maturity result.** Short-term bank debt creates hold-up in good states; long-term bank debt creates bribe-like distortions in bad states. Optimal maturity depends on bargaining power.
- **Source result.** High-quality firms tilt toward arm's-length debt; weaker firms can prefer bank finance because control is more valuable.
- **Capital-structure result.** Firms can use multiple lenders and priority design to preserve bank control while limiting bank rents.
- **Dynamic result.** As credible public information arrives, arm's-length debt becomes more attractive relative to bank debt.

8 Conclusion

8.1 Core conclusion (what the paper establishes)

The paper establishes that informed lending is not unambiguously superior to arm's-length lending. Banks solve an important control problem, but they also create an incentive problem because their informational advantage lets them bargain away part of the owner's surplus once the project is under way.

8.2 Economic intuition

The choice between bank debt and market debt is really a choice about who gets control when future contingencies are unverifiable. If outsiders finance the firm, the owner keeps more upside but can inefficiently continue bad projects. If a bank finances the firm, the bank can stop those mistakes, but it becomes a powerful insider. The owner then works less hard ex ante because she knows some of the gains will be appropriated later.

8.3 Takeaway

Rajan's contribution is to show that debt structure is a way of allocating informational power. Source, maturity, and priority are all tools for deciding how much control an informed lender should have and how much of that lender's ex post rent extraction the firm is willing to tolerate.